How the MES4-Carrier-EC works

24 V 5 V signals and buses

+24V rail: relay contacts, INA226 VBUS and sensor supplies

24 V DC in (J5, XT30)

XT30 → F1 15A fuse → Q1 reverse-polarity P-FET → D9 SMBJ30A TVS + C1 100 µF bulk → +24V rail.
LED3 = 24 V present.

+24V

Filter + 24 V → 5 V (U1)

L1 pi filter → URB2405S-6WR3 (isolated, 1.2 A) → +5V_24: relay coils and the module feed.

+5V_24

Module 5 V feed

F2 1.1 A PTC + FB1 ferrite pi → +5V: module fingers 20 / 22 and the ADS1232 excitation.

+5V

MES4-MCU module (J1, M.2 socket)

STM32G0B1, 64 MHz, CAN transceiver.
Gets +5V on fingers 20 / 22.
Makes its own 3.3 V → +3V3_MOD (finger 4, ≤ 50 mA) for the I²C pull-ups, INA226s, PC817 outputs and the ADS1232 logic.

All 14 GPIOs are used (GPIO map on the Main sheet).

LED1 = module +5V, LED2 = 3V3.
Hold-down: H1 M3 standoff.

No hot-plug: unplug the XT30 before fitting a module.

PA8 / PC6 / PB3 / PA7

coil FB PA0-PA3

I²C1 + PA15

PB6 / PC15

PA4-PA6

SWD, NRST, VCP

CAN

4 relay outputs (CIF §7)

1 safety light: PA8 drive, PA0 FB, INA226 0x40
2 LED DRL: PC6 drive, PA1 FB, 0x41
3 battery stopper: PB3 drive, PA2 FB, 0x44
4 spare: PA7 drive, PA3 FB, 0x45

GPIO → 100 Ω → AO3400A → 5 V coil (SRD-5VDC-SL-A, on +5V_24).
+24V → 30 mΩ shunt → contact → 3A PTC → output (2 A continuous).
Coil feedback 10k / 15k → PA0-PA3.
INA226 per channel: load current + 24 V rail, ALERT → PA15.

2 Autonics sensor inputs

+24V → 100 mA PTC → sensor supply.
Sensor OUT → 10k → PC817 → PB6 / PC15 (HIGH = sensor ON). PNP 24 V sensors.

Load cell (U8 ADS1232)

4-wire bridge, excited from +5V (ratiometric, as on the MES4-Flasher).
100 Ω / 100 nF / 10 nF filters, BAV199 clamps. 24 bit, gain 128, 10 SPS.
DRDY PA4, SCLK PA5, PDWN PA6.

Winch control (page 10)

Two IM03GR relays, as on the MES2_Extension (AO3400A drivers).
TODO: drive from a TCA9535 on I²C1 (PA0 / PA1 are coil FB).
Contacts switch the 48 V winch controller's direction inputs.
Hardware interlock: both on = both open.

IMU 9 V + screen 5 V (page 9)

Own L pi filter from +24V.
U9 URB2409S-10WR3 → 9 V → SW2 switch → J14 IMU terminal (as on the MES3.1 carrier).
U10 URB2405LD-20WR3 → 5 V 4 A → 3 A PTC → J16 USB-C (Rp 10k: 5 V / 3 A) → ARZOPA Z1C screen.
Door servo (FE-URT-2): J15 24 V + J17 UART via the I²C bridge.

ST-Link + reset (J2, SW1)

STDC14: SWD, NRST, VCP.
SW1 pulls NRST low.
H5VSD3B ESD on every pin.

CAN (J3 DB9, J4 RJ45 x4)

DB9 male to the IPC (1 GND, 3 H, 5 L) + 1x4 RJ45 (1 H, 2 L, 3 GND), one bus.
H24VNT2B ESD at each, no series R.

Notes

E-Crawler carrier for one MES4-MCU module (MES4-CIF rev A-d6). Circuits taken from the MES4-Flasher: PC817 sensor inputs, ADS1232 load cell, relay channels (driver, coil feedback, INA226) and the XT30 / URB2405S 24 V input.
GPIO budget: 4 relay drives + 4 coil feedbacks + ALERT + 3 ADS1232 + 2 sensors = 14, as in the MES4-CIF §2c table.
Servo UART (SC16IS740) and, next, the winch (TCA9535) run on I²C1, so they need no module pins.
Relay drive per CIF §7a (AO3400A, 100 Ω, 100k pull-down); the Flasher uses a 2SC2884 BJT for the same job.
No hot-plug: unplug the 24 V XT30 before inserting or removing the module.
Pages: Main (2), CAN (3), Power (4), Relays 1-2 (5), Relays 3-4 (6), Sensors (7), LoadCell (8), AuxPower (9), Winch (10).

Pages

Main	CAN	Power	Relays1-2	Relays3-4
File: Main.kicad_sch	File: CAN.kicad_sch	File: Power.kicad_sch	File: Relays12.kicad_sch	File: Relays34.kicad_sch

Sensors	LoadCell	AuxPower	Winch
File: Sensors.kicad_sch	File: LoadCell.kicad_sch	File: AuxPower.kicad_sch	File: Winch.kicad_sch

E-Crawler carrier for the MES4-MCU M.2 module
Aaron Raj Bhatti
Monkey Robotics

Sheet: /
File: MES4_Carrier_EC.kicad_sch

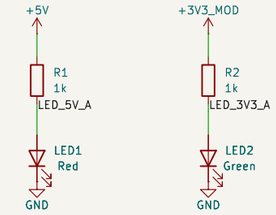
Title: MES4-Carrier-EC

Size: A3	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 9.0.2		Id: 1/10

Module GPIO map: all 14 used

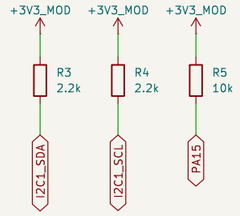
PA8 finger 31 out relay 1 drive, safety light (Q2)
PC6 finger 29 out relay 2 drive, LED DRL (Q3)
PB3 finger 61 out relay 3 drive, battery stopper (Q4)
PA7 finger 35 out relay 4 drive, spare (Q5)
PA0 finger 49 in   relay 1 coil feedback (10k / 15k divider + 10 nF)
PA1 finger 47 in   relay 2 coil feedback
PA2 finger 45 in   relay 3 coil feedback
PA3 finger 43 in   relay 4 coil feedback
PA15 finger 63 in   INA226 ALERT, all four wire-OR (R5 10k pull-up)
PA4 finger 41 in   ADS1232 DRDY/DOUT (U8)
PA5 finger 39 out ADS1232 SCLK (100k pull-down)
PA6 finger 37 out ADS1232 PDWN (100k pull-down)
PB6 finger 59 in   Autonics sensor 1 (U6 PC817)
PC15 finger 51 in   Autonics sensor 2 (U7 PC817)
I²C1 SDA / SCL fingers 53 / 55: INA226 0x40 / 0x41 / 0x44 / 0x45, SC16IS740 0x48 (servo UART)
Not used: USB 23 / 25 (open, CIF rule 8). VCP TX / RX: STDC14 only.
Matches the MES4-CIF §2c GPIO table (the MES4 standard). The winch and servo use I²C1,
not module pins. TODO: winch on TCA9535 outputs (HANDOFF.md step 2).

Power indicators



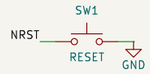
LED1 red = module +5V present (3 mA).
LED2 green = module 3V3 present
(fed from finger 4, ~0.7 mA).

I²C1 + ALERT pull-ups



CIF rule 7: always fitted,
to the module's 3V3 (finger 4).
2.2k: 4 × INA226 at 400 kHz.
R5: INA226 ALERT wire-OR (PA15).

Reset



Pulls NRST low only.
No capacitor: C6 100 nF
is on the module (CIF §4a).

CAN bus (page 3)

Module CANH / CANL (fingers 1 / 3) go to the CAN sheet
as global labels:

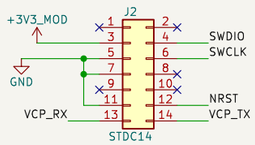
J3 DB9 male to the IPC: 1 GND, 3 CAN_H, 5 CAN_L.
J4 1x4 RJ45, no magnetics: 1 CAN_H, 2 CAN_L,
3 GND on every port (T568B: white-orange, orange,
white-green).

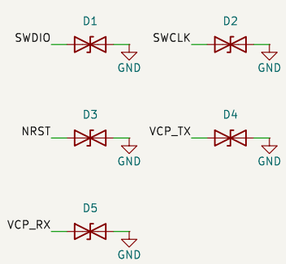
All five ports are one bus, wired in parallel.
ESD at each connector: D6, D7 (H24VNT2B).
Transceiver + 120 Ω terminator (JP1) are on the module.

Links to the other pages

Page 1: block diagram and page list.
Page 3 CAN: DB9 male to the IPC + 1x4 RJ45, ESD.
Page 4 Power: XT30 24 V input, fuse, reverse-polarity FET, TVS, noise filter, 24 V → 5 V.
Pages 5–6 Relays: PA8 / PC6 / PB3 / PA7 drive, PA0–PA3 coil feedback, 4 × INA226.
Page 7 Sensors: PB6 / PC15 (two Autonics PNP sensors).
Page 8 LoadCell: PA4 DRDY, PA5 SCLK, PA6 PDWN.
Page 9 AuxPower: IMU 9 V (MES3.1), screen USB-C 5 V / 3 A, door servo (FE-URT-2, UART via the I²C bridge).
Page 10 Winch: CLK / ACLK direction relays (TODO: TCA9535 drive) + SC16IS740 servo UART bridge (I²C 0x48).
Signals shared between pages are global labels.

ST-Link V3MINIE (STDC14) + ESD

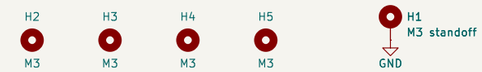




STDC14 per ST UM2448 Table 6: 3 T_VCC, 4 SWDIO, 6 SWCLK, 12 NRST,
13 T_VCP_RX (probe → PA10), 14 T_VCP_TX (probe ← PA9), 5/7/11 GND.
Never tie SWD, NRST or VCP to GND (MES3 bug). No BOOT0 button (CIF §4).

D1–D5: H5VSD3B ESD (5 V, 1.5 pF), pin 2 to GND.

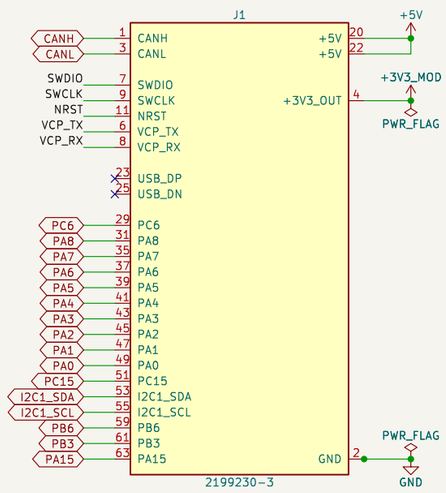
Mechanical



Design notes

1. Purpose: E-Crawler carrier for one MES4-MCU module (follows MES4-CIF rev A-d6). Circuits from the MES4-Flasher: relays, opto inputs, load cell, 24 V in.
2. No hot-plug (rule 5): unplug the 24 V XT30 before inserting or removing a module (no power switch).
3. +5V (fingers 20 / 22) = U1 URB2405S → F2 PTC → FB1 ferrite: 4.75–5.5 V, ≥ 300 mA (rule 2, 4).
4. +3V3_MOD = module LDO output on finger 4, never driven (rule 3, ≤ 50 mA): LED 0.7 + pull-ups ≤ 3.3 + 2 × PC817 0.7 + 4 × INA226 1.7 + ADS1232 0.1 + ST-Link VTref ≈ 10 mA.
5. NRST: SW1 and STDC14 pin 12 only pull low (wired-OR). No extra capacitance (C6 on the module).
6. ESD (rule 8): H5VSD3B on every STDC14 signal, H24VNT2B on CANH / CANL, no series R on CAN.
7. No back-powering (rule 6): carrier logic runs from +3V3_MOD (pull-ups, PC817s, INA226, ADS1232 DVDD).
8. All relay drives OFF in reset: 100k gate pull-downs.
9. Parts: LCSC numbers in each symbol's LCSC field.

MES4-MCU module socket (TE 2199230-3, M.2 B-key 4.2 mm)



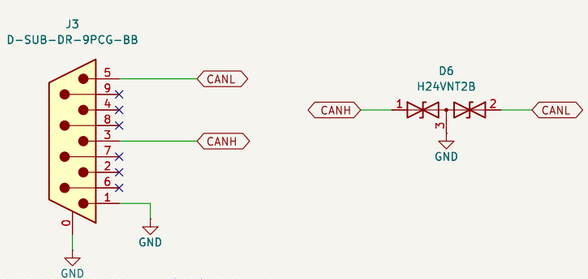
Odd fingers (card top) on the left, power on the right.
+5V = fingers 20 + 22. GND = all other fingers + the
hold-down pads (stacked pin). USB 23/25 open (CIF rule 8).
Module hold-down: H1 M3 × 2.5 mm SMD standoff + M3
screw, 40.25 mm from the socket posts (42 mm card).

Board connectors

J5 24V IN, XT30 male right-angle: 1 = +24 V (flat end),
2 = GND_24V (chamfered end).
J6 / J7 / J8 / J9 RLY1–4 OUT (KF250NH 2-pin):
1 = switched +24 V, 2 = GND_24V.
J10 / J11 SENS1 / SENS2 (KF250NH 3-pin): 1 = +24 V
(100 mA PTC, brown), 2 = output (black), 3 = 0 V (blue).
J12 LC SIG: 1 SIG+ green, 2 SIG- white, 3 shield.
J13 LC EXC: 1 EXC+ red, 2 EXC- black, 3 shield.
5 V load cell only: never 24 V.
J3 CAN DB9 male (IPC): 1 GND, 3 CAN_H, 5 CAN_L.
J4 CAN RJ45 x4: 1 CAN_H, 2 CAN_L, 3 GND (T568B).
J14 IMU 9V (KF250NH 2-pin): 1 = +9 V after the switch, 2 = GND.
J16 SCREEN USB-C: 5 V / 3 A power only (no data).
J15 SERVO 24V: 1 GND_24V, 2 +24 V (3 A PTC).
J17 SERVO UART: 1 GND, 2 +5 V, 3 RX, 4 TX.
J18 WINCH: 1 CLK, 2 COM, 3 ACLK (relay contacts).
J2 STDC14: ST-Link V3MINIE (SWD, NRST, VCP).
Test points: Power sheet.

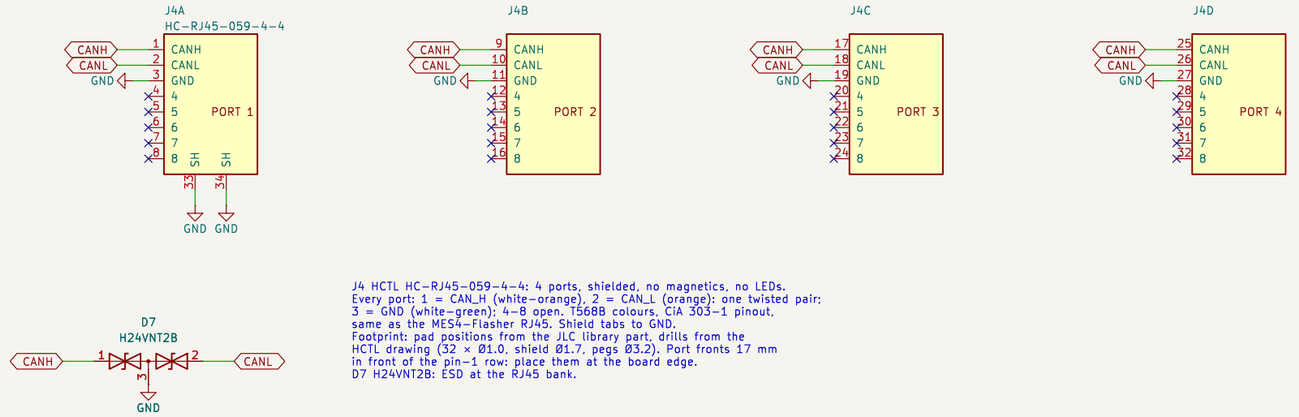
Sheet 2: module socket, ST-Link, reset, I²C pull-ups, power LEDs
Aaron Raj Bhatti
Monkey Robotics
Sheet: /Main/
File: Main.kicad_sch
Title: MES4-Carrier-EC
Size: A3 | Date: 2026-10-27 | **Rev: A**
KiCad E.D.A. 9.0.2 | Id: 2/10

IPC link: DB9 male



J3 XUNPU D-SUB-DR-9PCG-BB: DB9 male (pins), right angle.
4-40 shell, KiCad DSUB-9 footprint (EdgePinOffset 7.70 mm).
Pinout per the IPC: 1 = GND, 3 = CAN_H, 5 = CAN_L, the rest open.
Shell / mounting holes to GND (as the RJ45 shield).
Male on the board and on the IPC: use a straight-through
female-female DB9 cable (1-1, 3-3, 5-5).
Not the CiA 303-1 DB9 pinout (2 CAN_L, 7 CAN_H, 3 GND):
a standard CAN DB9 cable or adapter will not match.
D6 H24VNT2B: ESD at the connector, no series R (CIF rule 8).

CAN ports: 1x4 RJ45, no magnetics



J4 HCTL HC-RJ45-059-4-4: 4 ports, shielded, no magnetics, no LEDs.
Every port: 1 = CAN_H (white-orange), 2 = CAN_L (orange): one twisted pair;
3 = GND (white-green); 4-8 open. T568B colours, CiA 303-1 pinout,
same as the MES4-Flasher RJ45. Shield tabs to GND.
Footprint: pad positions from the JLC library part, drills from the
HCTL drawing (32 × Ø1.0, shield Ø1.7, pegs Ø3.2). Port fronts 17 mm
in front of the pin-1 row: place them at the board edge.
D7 H24VNT2B: ESD at the RJ45 bank.

CAN bus rules (CIF §5, rules 8-9)

All five ports (DB9 + 4 × RJ45) are the same bus, wired in parallel to the module's CANH / CANL (fingers 1 / 3).
Keep the bus a line: IPC → DB9 → this board → one RJ45 → next node → ... A port used as a branch is a stub:
keep each stub ≤ 0.3 m at 500 kbit/s. More long branches need a hub with its own transceivers, not more ports.
Exactly two 120 Ω terminators, at the two physical ends of the bus: bridge JP1 on the module only if this board is an end.
All nodes on one solid ground: pin 3 (RJ45) / pin 1 (DB9) carry it. Check with power off: CAN_H-CAN_L ≈ 60 Ω.
Never carry 24 V in a CAN cable or connector (CIF rule 9): these jacks carry only CAN_H, CAN_L and GND.
No series resistors on CAN_H / CAN_L (CIF rule 8): the module must drive the 60 Ω bus directly.

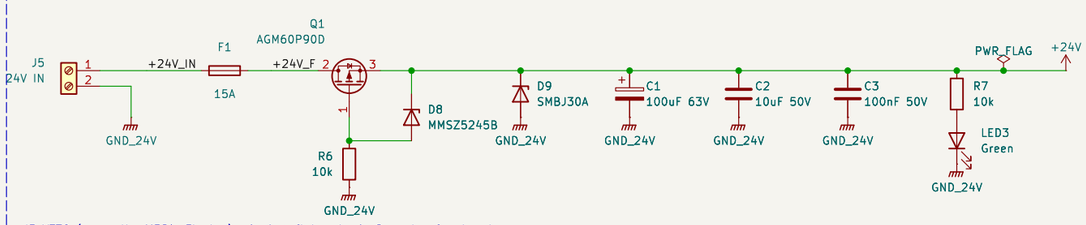
Sheet 3: CAN bus connectors: DB9 male to the IPC + 1x4 RJ45 (no magnetics), ESD
Aaron Raj Bhatti
Monkey Robotics

Sheet: /CAN/
File: CAN.kicad_sch

Title: MES4-Carrier-EC

Size: A3	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 9.0.2		Id: 3/10

24 V input: XT30, fuse, reverse-polarity FET, TVS, bulk



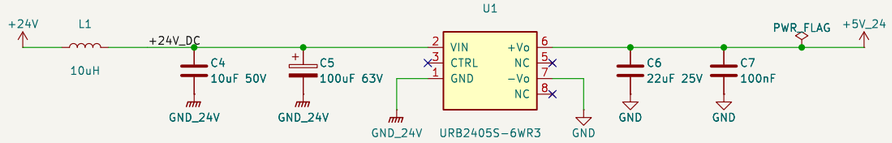
J5 XT30 (as on the MES4-Flasher): pin 1 + flat end, pin 2 − chamfered end.
F1 15A fast fuse (Nano2 2410, 65 V): protects the XT30 and the board copper; the 8.5 A worst case stays under its 75 % derating. Each relay output has its own 3 A PTC, so a load short never blows F1.
Q1 P-MOSFET reverse-polarity block: correct polarity turns it on (Vgs clamped by D8, R6 10k to GND_24V); a reversed lead leaves it off. It replaces the Flasher's SS56 Schottky, which would dissipate ≈ 3–4 W here. AGM60P90D: −60 V, 13 mΩ, ≈ 1 W at 8.5 A: copper on the drain tab.
D9 SMBJ30A (600 W): surges and spikes from motors and solenoids on the 24 V bus (CIF §7a).
C1 100 µF / 63 V electrolytic: bulk, and its ESR damps the cable inductance (hot-plug ringing).
C2 / C3: HF bypass. LED3 green = 24 V present (after Q1).
+24V feeds the relay contacts, the INA226 VBUS inputs and the sensor supplies.

Power budget and noise strategy

24 V current: relays ≤ 4 × 2 A (CIF §7; realistic 3–5 A: lamp ~0.3 A, DRL, 1.5 A solenoid), sensors 2 × ≤ 100 mA, logic DC/DC ≈ 0.15 A, IMU 9 V ≤ 0.5 A + screen 5 V ≤ 0.95 A + door servo ≤ 3 A (page 9). Realistic total ≈ 6–8 A. Every load at its limit at once (≈ 13 A) would exceed F1 15A's 75 % derating (11 A) and the XT30's 15 A rating: don't stall the servo while all four relays carry 2 A.

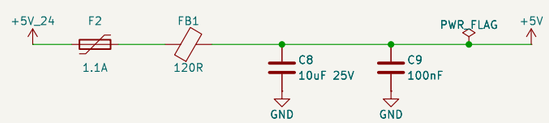
Noise and transients, from the cable inwards:
1. D9 SMBJ30A clamps surges and inductive spikes on the 24 V bus.
2. C1 electrolytic damps the cable LC: no ringing above 24 V on hot-plug.
3. C2 / C3 ceramics: low impedance for HF at the input.
4. Pi filter C2 + L1 + C4 / C5 feeds only the DC/DC: the logic supply is decoupled from the bus, and the converter's switching noise stays off it.
5. Isolated DC/DC: the grounds meet only at NT1, so relay load currents never flow in GND.
6. Pi filter F2 + FB1 + C8 / C9 on the module 5 V, and FB2 again on the ADS1232 AVDD.
7. Flyback diodes at every relay coil (1N4148W) and every output (SS34).
8. Sensor supplies: 100 mA PTC + 100 nF / 50 V at each terminal.

Noise filter + isolated 24 V → 5 V



L1 10 µH + C4 / C5, with C2 on +24V: pi filter in front of the DC/DC only.
Relay contacts, INA226s and sensors use the unfiltered +24V.
U1 URB2405S-6WR3 (as on the MES4-Flasher): 9–36 V in, 5 V 1.2 A, short-circuit protected.
CTRL open = on. C5 = the maker's 100 µF input cap; C6 / C7 = output caps.
The maker's EMC circuit adds a 470 µH common-mode choke + 1 nF Y caps: not fitted, because NT1 ties the grounds (the 1.5 kV isolation is not used). The maker lists hot-plug as unsupported: L1 slows the inrush into C5; Q1 / D9 / C1 take the XT30 plug-in.
Load: module ≤ 240 mA + 4 coils 285 mA + 2 winch coils 56 mA + load cell 13 mA ≈ 0.6 A (50 %).

Module + ADC 5 V feed (CIF rule 4)



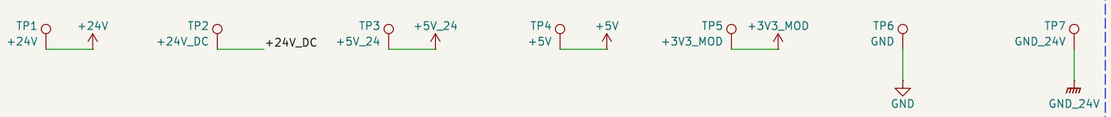
F2 1.1 A PTC: the fuse per module slot (CIF rule 4).
FB1 + C8 / C9: pi filter, so relay-coil switching on +5V_24 stays off the module and the load-cell excitation.
+5V → module fingers 20 / 22 (4.75–5.5 V, CIF rule 2) and the ADS1232 AVDD (LoadCell sheet). Place C8 / C9 at the socket.

Grounds



GND_24V: 24 V return (XT30 pin 2, relay outputs, sensor 0 V, TVS, bulk caps, DC/DC input side).
GND: logic (module, INA226, ADS1232, DC/DC output).
NT1: the only joint between them (the INA226s measure +24V against GND). Place it at the DC/DC.

Test points (bring-up)



2 mm pads with a 1 mm hole: probe tip, or solder a wire loop for clips (not in the BOM).
Clip the scope ground to GND for the module side and to GND_24V for the 24 V side.

First power-up

1. No module fitted. Apply 24 V on J5 from a current-limited supply: LED3 lights; +24V, +24V_DC, +5V_24 and +5V present (LED1 on).
2. Reverse the lead briefly: nothing lights and +24V stays at 0 V (Q1 blocks).
3. Unplug 24 V, fit the module, plug in again: LED2 lights (module 3V3). Then flash over STDC14 (J2).

Sheet 4: 24 V input + noise filtering, isolated 24 V → 5 V, module 5 V feed
Aaron Raj Bhatti
Monkey Robotics
Sheet: /Power/
File: Power.kicad_sch
Title: MES4-Carrier-EC
Size: A3 | Date: 2026-10-27 | **Rev: A**
KiCad E.D.A. 9.0.2 | Id: 4/10

Relay diagnosis (CIF §7b)

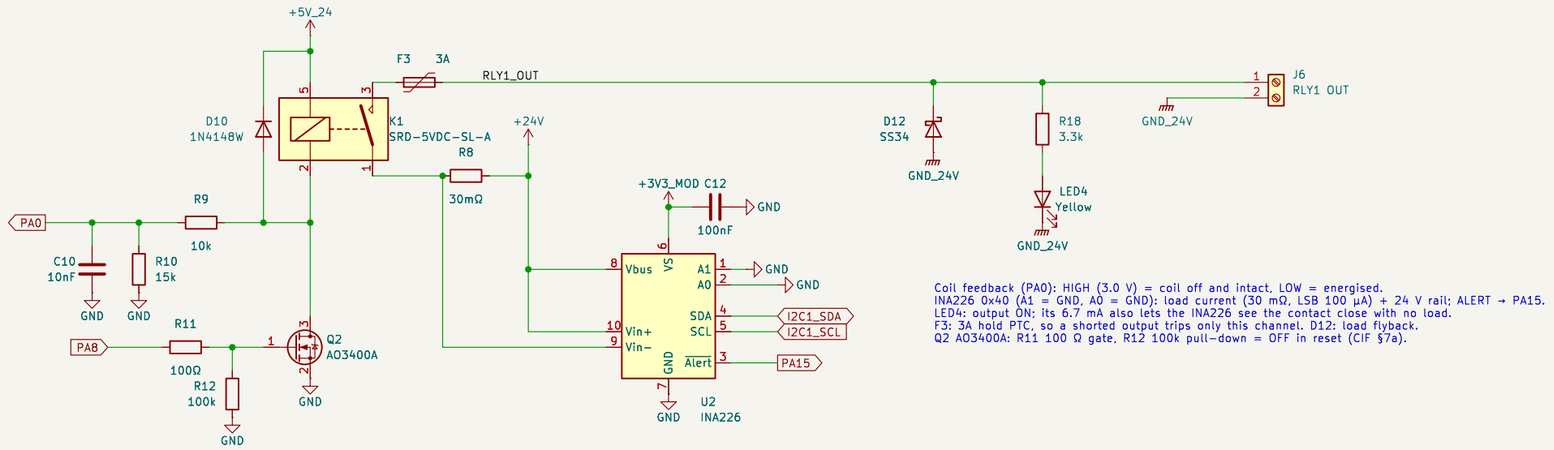
Coil supply = +5V_24 (U1 DC/DC, Power sheet).
Read ≥ 30 ms after switching.

OFF, FB HIGH, I ≈ 0 → OK, relay off
OFF, FB LOW → open coil, no +5V_24
   or shorted driver
OFF, FB HIGH, I > 5 mA → contacts welded
ON, FB LOW, I in window → OK (LED 6.7 mA + load)
ON, FB HIGH → driver or GPIO not switching
ON, FB LOW, I ≈ 0 → contact not closing,
   output PTC or input fuse open, or no 24 V
ON, I over limit → short / stall (ALERT → PA15)
any, VBUS < 20 V → 24 V supply fault

FB = coil feedback: 10k / 15k divider + 10 nF into PA0–PA3 (CH1–CH4, CIF §2c): 3.0 V when the coil is off and intact, ~0 V when energised, ≈ 3.5 V at the turn-off peak: fine on PA0–PA3 (5 V–tolerant), never move it to PA4 / PA5. ADC–capable: firmware can also read the voltage. I = INA226 load current, VBUS = 24 V rail.

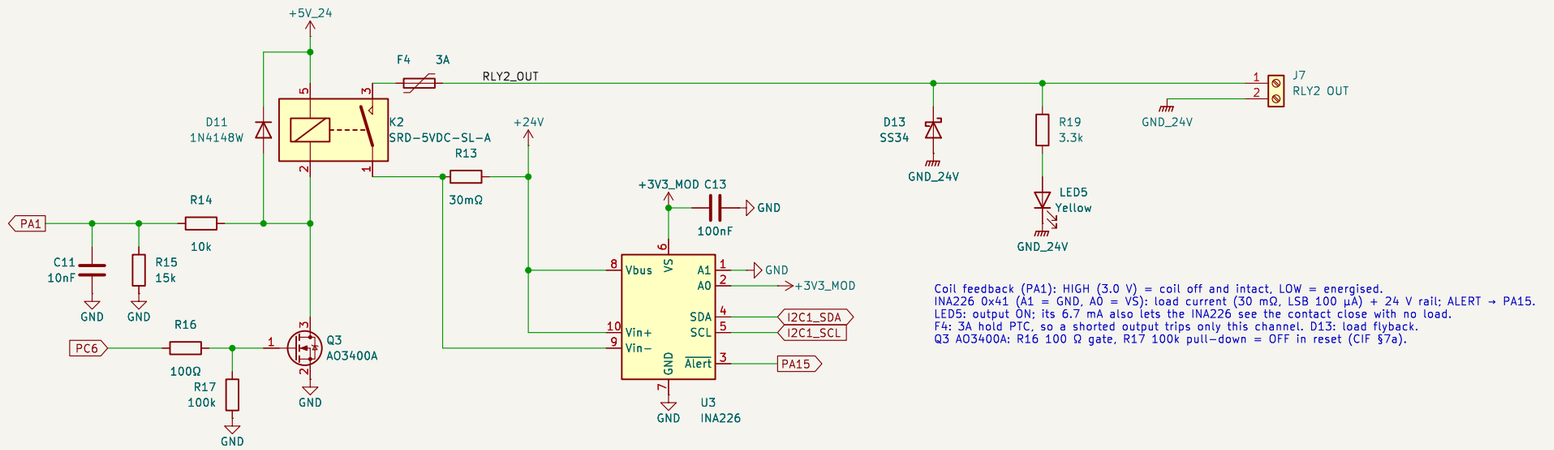
INA226 (30 mΩ, CIF §6a): CONFIG 0x4327, CAL 1707 (LSB 100 µA, FS 2.73 A), MASK/ENABLE 0x8001 (SOL, latched), ALERT limit 30000 = 2.5 A default; set each channel's value from the CANopen dictionary. "Current present" threshold 5 mA. On ALERT LOW: read MASK/ENABLE of each INA226 (AFF bit 4) to find the channel; the read also clears the latch. Switch it OFF, report on CAN.

Relay 1 (safety light): drive PA8, coil feedback PA0, INA226 U2 at 0x40



Coil feedback (PA0): HIGH (3.0 V) = coil off and intact, LOW = energised.
INA226 0x40 (A1 = GND, A0 = GND): load current (30 mΩ, LSB 100 µA) + 24 V rail; ALERT → PA15.
LED4: output ON; its 6.7 mA also lets the INA226 see the contact close with no load.
F3: 3A hold PTC, so a shorted output trips only this channel. D12: load flyback.
Q2 AO3400A: R11 100 Ω gate, R12 100k pull–down = OFF in reset (CIF §7a).

Relay 2 (LED DRL): drive PC6, coil feedback PA1, INA226 U3 at 0x41



Coil feedback (PA1): HIGH (3.0 V) = coil off and intact, LOW = energised.
INA226 0x41 (A1 = GND, A0 = VS): load current (30 mΩ, LSB 100 µA) + 24 V rail; ALERT → PA15.
LED5: output ON; its 6.7 mA also lets the INA226 see the contact close with no load.
F4: 3A hold PTC, so a shorted output trips only this channel. D13: load flyback.
Q3 AO3400A: R16 100 Ω gate, R17 100k pull–down = OFF in reset (CIF §7a).

Sheet 5: relay outputs 1–2 (safety light, LED DRL) with coil and load–current feedback
Aaron Raj Bhatti
Monkey Robotics

Sheet: /Relays1–2/		
File: Relays12.kicad_sch		
Title: MES4–Carrier–EC		
Size: A3	Date: 2026–10–27	**Rev: A**
KiCad E.D.A. 9.0.2		Id: 5/10

Loads (CIF §7, all 24 V DC)

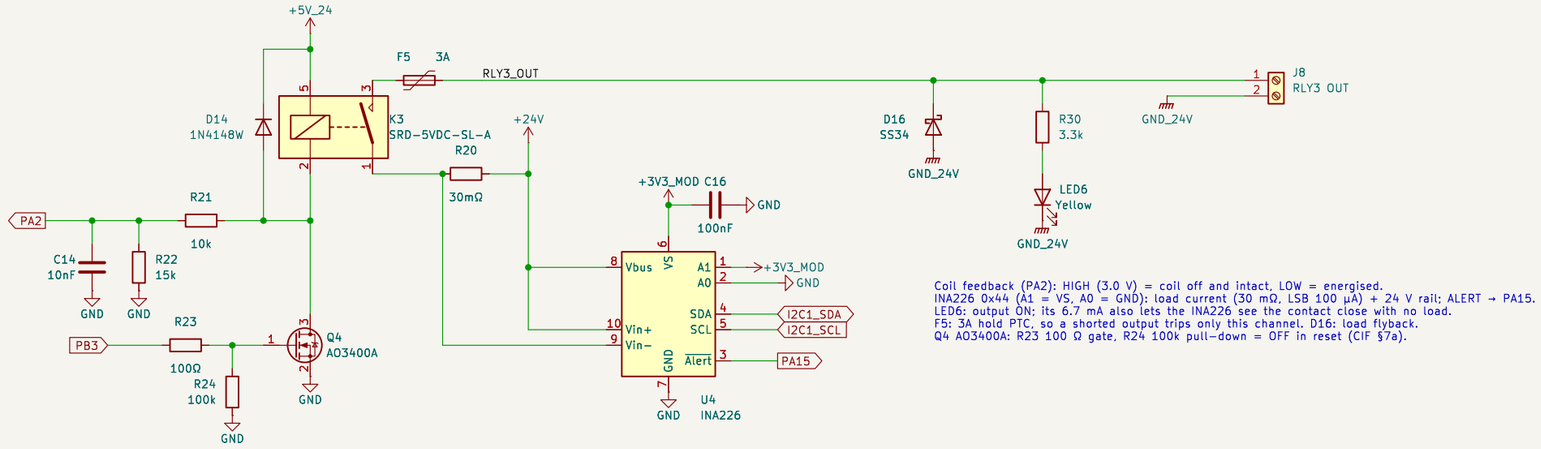
CH1 safety light: Hanyoung T084-PAD24Y-D
(LED + motor). Current TBD: measure,
then set its window (est. 0.1–0.3 A).
CH2 LED DRL: ≤ 2 A continuous.
CH3 battery stopper: Heschen HS-1564B 24 V
push-pull solenoid, 1.5 A, intermittent
(80 °C after 3 min continuous): firmware
enforces a maximum on-time + cool-down.
CH4 spare: ≤ 2 A continuous.

Every channel: 24 V / 2 A continuous,
3A hold PTC per output (CIF §7a: fuse
per channel). Shunt on the supply side of the
contact, so the INA226 never sees the load
flyback (−0.7 V < −0.3 V input limit) and a
welded contact still shows current while OFF.

Inductive loads (solenoid): SS34 flyback at
each output on the board; also fit an SS14-class
diode at the solenoid itself.

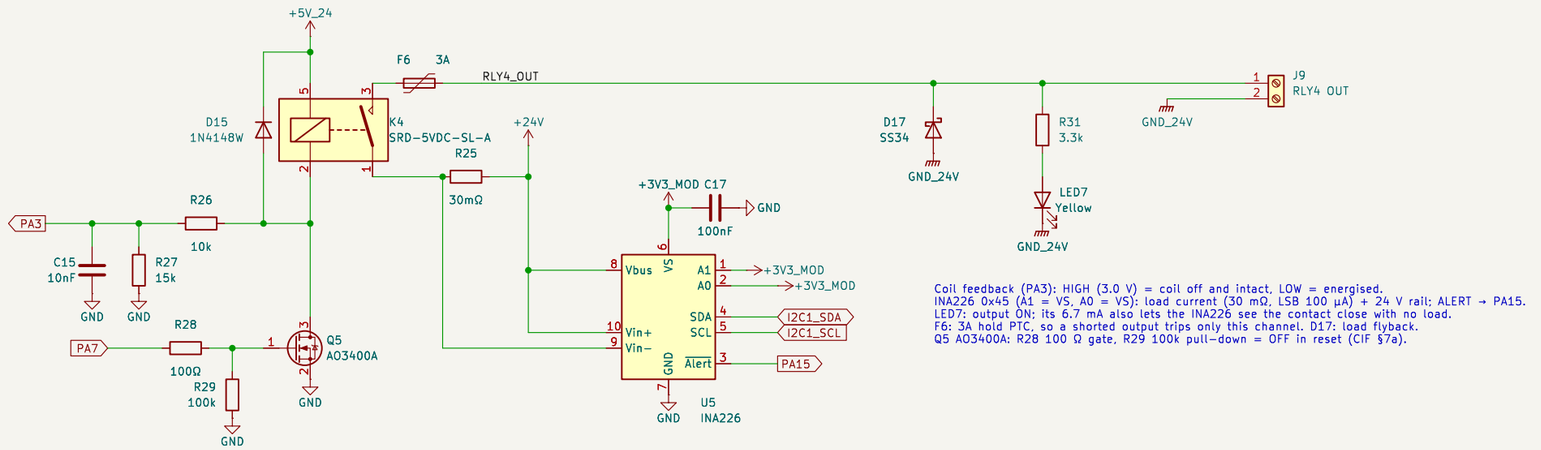
Coil budget: 4 × 71 mA ≈ 285 mA from +5V_24
(U1 URB2405S, 1.2 A).
All relays OFF in reset / boot / firmware
update: 100k gate pull-downs; PA8 reads LOW in
reset (UCPD pull-down). PA7 is pulled DOWN by
the bootloader (CIF §2c).

Relay 3 (battery stopper): drive PB3, coil feedback PA2, INA226 U4 at 0x44



Coil feedback (PA2): HIGH (3.0 V) = coil off and intact, LOW = energised.
INA226 0x44 (A1 = VS, A0 = GND): load current (30 mΩ, LSB 100 µA) + 24 V rail; ALERT → PA15.
LED6: output ON; its 6.7 mA also lets the INA226 see the contact close with no load.
F5: 3A hold PTC, so a shorted output trips only this channel. D16: load flyback.
Q4 AO3400A: R23 100 Ω gate, R24 100k pull-down = OFF in reset (CIF §7a).

Relay 4 (spare): drive PA7, coil feedback PA3, INA226 U5 at 0x45

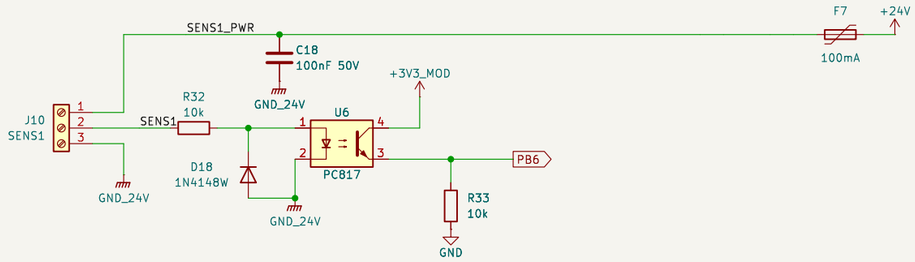


Coil feedback (PA3): HIGH (3.0 V) = coil off and intact, LOW = energised.
INA226 0x45 (A1 = VS, A0 = VS): load current (30 mΩ, LSB 100 µA) + 24 V rail; ALERT → PA15.
LED7: output ON; its 6.7 mA also lets the INA226 see the contact close with no load.
F6: 3A hold PTC, so a shorted output trips only this channel. D17: load flyback.
Q5 AO3400A: R28 100 Ω gate, R29 100k pull-down = OFF in reset (CIF §7a).

Sheet 6: relay outputs 3–4 (battery stopper, spare) with coil and load-current feedback
Aaron Raj Bhatti
Monkey Robotics

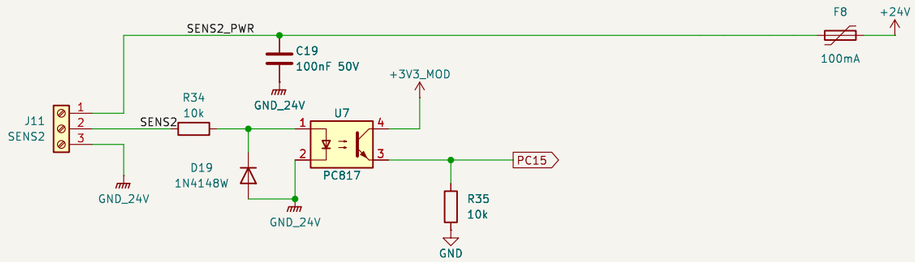
Sheet: /Relays3-4/		
File: Relays34.kicad_sch		
Title: MES4-Carrier-EC		
Size: A3	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 9.0.2		Id: 6/10

Sensor 1 → PB6: Autonics PNP proximity sensor, PC817 opto input



J10: 1 = +24 V (brown) through F7 100 mA PTC, 2 = sensor output (black), 3 = 0 V (blue).
Output ON (24 V): ≈ 2.3 mA through R32 10k + U6 LED → PB6 HIGH (3.3 V, emitter follower).
Output OFF, cable unplugged or broken: PB6 LOW (R33 10k pull-down).
D18 1N4148W: reverse clamp across the opto LED. C18 100 nF / 50 V: supply decoupling + ESD.
A shorted sensor cable trips F7 only: the other sensor and the board keep running.

Sensor 2 → PC15: Autonics PNP proximity sensor, PC817 opto input



J11: 1 = +24 V (brown) through F8 100 mA PTC, 2 = sensor output (black), 3 = 0 V (blue).
Output ON (24 V): ≈ 2.3 mA through R34 10k + U7 LED → PC15 HIGH (3.3 V, emitter follower).
Output OFF, cable unplugged or broken: PC15 LOW (R35 10k pull-down).
D19 1N4148W: reverse clamp across the opto LED. C19 100 nF / 50 V: supply decoupling + ESD.
A shorted sensor cable trips F8 only: the other sensor and the board keep running.

Sensor inputs: notes

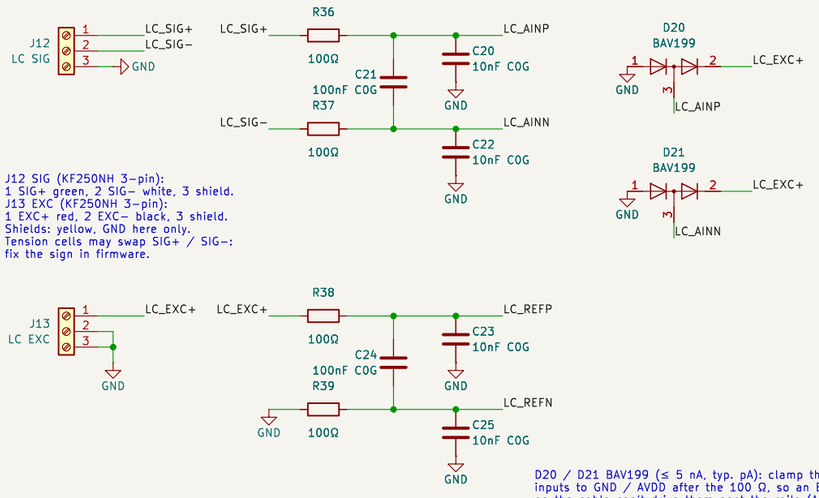
For 3-wire PNP (sourcing) 24 V sensors, e.g. Autonics PRD08 / PR series (brown +V, black OUT, blue 0 V).
NPN (sinking) sensors need a different input circuit.

Same circuit as the MES4-Flasher sensor channels 3 / 4 (PB6 / PC15), without the INA226 supply monitors.
GPIO budget: all 14 module pins are used (Main sheet, MES4-CIF §2c). More sensors need inputs on
I²C1, e.g. a TCA9535 expander (0x20-0x27), polled.

PB6: the ST bootloader enables a 25-55 kΩ pull-up on PB6 (blank chip only); with the 10k pull-down it
reads LOW, which is harmless for an input. PC15: input only (RTC-domain pin, CIF rule 10): fine here.
Interrupts: PB6 → EXTI6. PC15 shares EXTI15 with PA15 (INA226 ALERT): give EXTI15 to PC15 (sensor 2)
and poll ALERT, which the INA226 latches (MASK/ENABLE LEN = 1), so no fault is missed.
No back-powering: the PC817 collectors run from +3V3_MOD (module finger 4), so nothing drives the
module pins while it is unpowered.

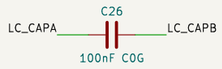
Sheet 7: two Autonics PNP sensor inputs (PC817 opto, as on the MES4-Flasher)
Aaron Raj Bhatti
Monkey Robotics
Sheet: /Sensors/
File: Sensors.kicad_sch
Title: MES4-Carrier-EC
Size: A3 | Date: 2026-10-27 | **Rev: A**
KiCad E.D.A. 9.0.2 | Id: 7/10

Load cell inputs J12 / J13 + RC filters (CIF §8a)



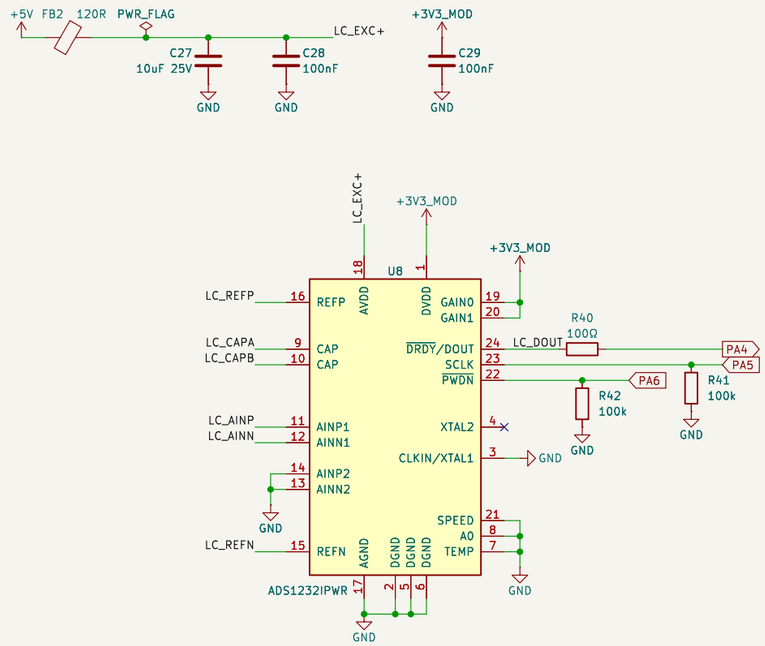
J12 SIG (KF250NH 3-pin):
1 SIG+ green, 2 SIG− white, 3 shield.
J13 EXC (KF250NH 3-pin):
1 EXC+ red, 2 EXC− black, 3 shield.
Shields: yellow, GND here only.
Tension cells may swap SIG+ / SIG−:
fix the sign in firmware.

D20 / D21 BAV199 (≤ 5 nA, typ. pA): clamp the ADC inputs to GND / AVDD after the 100 Ω, so an ESD hit on the cable can't drive them past the rails (ADS1232 is 750 V HBM). Leakage × ≈ 0.2 kΩ source ≪ 1 µV.



Signal and reference filters: 100 Ω per leg, 100 nF C0G across, 10 nF to GND (CIF §8a).
The reference gets the same network so both time constants match (TI SBAS350 §8.2).
REFP / REFN are sensed at J13 pins 1 / 2: route them as their own pair from the connector pads, not from the 5 V / GND planes. C26 = PGA bypass (pins 9–10).

U8 ADS1232 24-bit bridge ADC (TI SBAS350 weigh-scale configuration)



A0 = GND (channel 1). GAIN1 / GAIN0 = 1 / 1 → gain 128, ±19.5 mV at a 5 V reference. SPEED = GND → 10 SPS, rejects 50 and 60 Hz.
TEMP = GND. CLKIN = GND → internal oscillator, XTAL2 open. Unused channel 2 inputs to GND (TI weigh-scale example).
AVDD = +5V via FB2 = bridge excitation → ratiometric. AVDD range 2.7–5.3 V (SBAS350 §5.3).
PDWN (PA6) and SCLK (PA5) have 100k pull-downs: the ADC stays powered down until the firmware runs the power-up sequence.
DRDY/DOUT → PA4. PA4 / PA5 are 3.6 V-max pins: fine, all ADC logic runs at 3.3 V. Input common mode 2.5 V (needs 1.5–3.5 V at gain 128).

Load cell firmware (CIF §8b, TI SBAS350 §7.4.5)

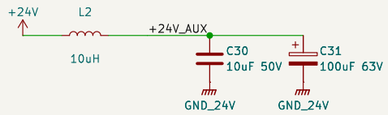
Power-up: keep PA6 (PDWN) LOW until +5V and 3V3 are stable, wait ≥ 10 µs, then PDWN HIGH ≥ 26 µs, LOW ≥ 26 µs, HIGH.
Read: wait for DRDY/DOUT LOW, clock 24 bits on SCLK (MSB first, two's complement). Clocking 26 bits instead also runs the offset self-calibration: do it after power-up and when the temperature changes. Discard the first conversions after power-up or reset.

Scale: 2 × SOP SOPLF-102 (1.5 t, 1.5 mV/V) summed in the junction box → 7.5 mV full scale at 5 V ≈ 38 % of ±19.5 mV, ≈ 2.5 µV per kg.
Bridge current ≈ 13 mA (two 780 Ω cells) from +5V. Warn at 1.5 t, alarm at 2.0 t total. Re-calibrate the scale factor (the old HX711 factor was taken at 3.3 V). Filtering as before: 0.5 kg noise threshold, ±0.25 kg stability window, 0.1 kg rounding.

5 V load cell only: J13 pin 1 is the +5V rail through FB2. Never land a 24 V wire on J12 / J13 (same push-in terminal type as the relay and sensor outputs): mark them "5V LOAD CELL" on the silk.
Excitation comes from +5V (Power sheet: URB2405S → PTC → ferrite pi filter), the same rail as the module.

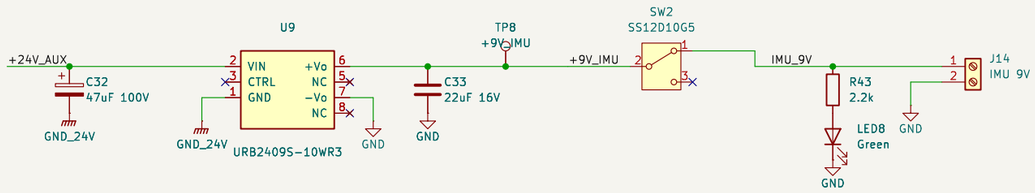
Sheet 8: load cell ADC (ADS1232, CIF §8, as on the MES4-Flasher)
Aaron Raj Bhatti
Monkey Robotics
Sheet: /LoadCell/
File: LoadCell.kicad_sch
Title: MES4-Carrier-EC
Size: A3 | Date: 2026-10-27 | **Rev: A**
KiCad E.D.A. 9.0.2 | Id: 8/10

Aux 24 V filter



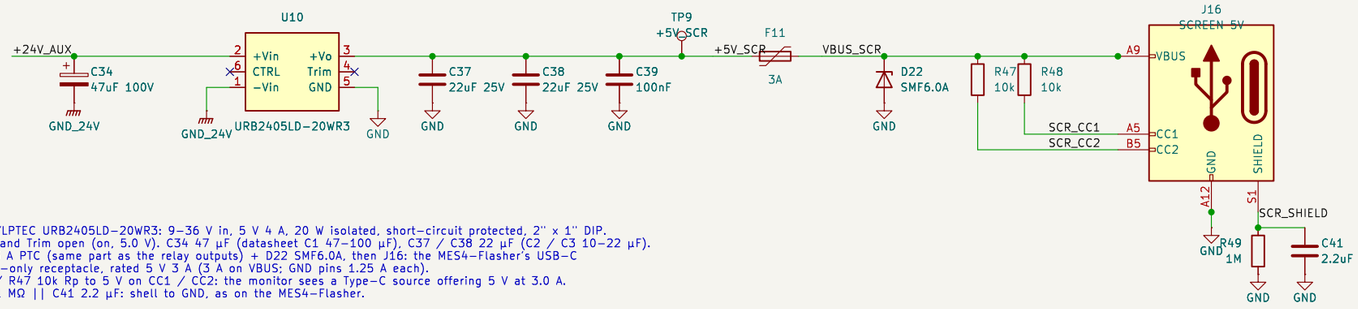
L2 10 µH (same part as L1) + C30 / C31: a pi filter
of its own for the IMU and screen converters, so the screen's
20 W converter neither loads nor disturbs the logic supply
(L1, Power sheet). Load at 24 V: IMU ≤ 0.5 A + screen ≤ 0.95 A
≈ 1.45 A; L2 is rated 2.1 A (2.9 A saturation).
Both converters are isolated: input side on GND_24V,
output side on GND (same scheme as the logic DC/DC).

IMU power: 9 V (as on the MES3.1 carrier)



Same parts as the MES3.1−Carrier−LH "IMU Power Supply @ 9V":
U9 YLPTEC URB2409S−10WR3 (9–36 V in, 9 V 1.11 A, 10 W isolated, short−circuit protected;
MES3.1 U15). C32 47 µF / 100 V and C33 22 µF / 16 V (the datasheet's Cin / Cout).
SW2 SS12D10G5 slide switch (MES3.1 U22, common = pin 2). J14 KF250NH 2−pin (MES3.1 CN10).
CTRL open = on. SW2 is rated 0.5 A: check the IMU draws less (the converter gives up to 1.11 A).
LED8 green (+ R43 2.2k, 3.2 mA) sits after SW2: it lights only when the IMU terminal is powered.
LCSC stock of URB2409S−10WR3 is low (78 on 2026−10−27): order early.

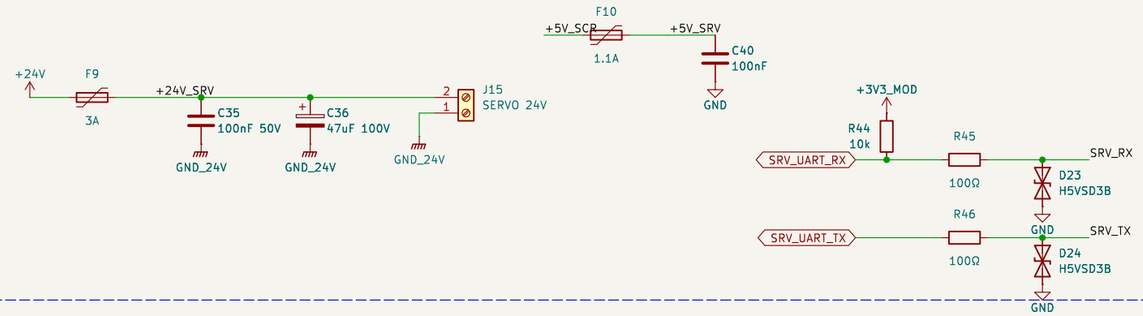
Screen power: USB−C 5 V / 3 A (ARZOPA Z1C portable monitor)



U10 YLPTEC URB2405LD−20WR3: 9–36 V in, 5 V 4 A, 20 W isolated, short−circuit protected, 2" x 1" DIP.
CTRL and Trim open (on, 5.0 V). C34 47 µF (datasheet C1 47–100 µF), C37 / C38 22 µF (C2 / C3 10–22 µF).
F11 3 A PTC (same part as the relay outputs) + D22 SMF6.0A, then J16: the MES4−Flasher's USB−C
power−only receptacle, rated 5 V 3 A (3 A on VBUS; GND pins 1.25 A each).
R48 / R47 10k Rp to 5 V on CC1 / CC2: the monitor sees a Type−C source offering 5 V at 3.0 A.
R49 1 MΩ || C41 2.2 µF: shell to GND, as on the MES4−Flasher.

ARZOPA Z1C label: input 5 / 9 / 12 / 15 / 20 V DC, 3 A max: a USB−PD sink.
This port has no PD controller, so it offers 5 V only (15 W max).
The sibling Z1RC draws about 9 W (≈ 1.8 A at 5 V), so 5 V / 3 A should
run it: check full brightness on the bench. If it needs 9 V or more,
the port needs a USB−PD source controller instead.
VBUS is always on (no cold−socket switch): plug in the monitor, or
another sink, only; never a charger or a laptop.
Power only: the video goes over the monitor's own HDMI / USB−C
from the IPC. Rp is to 5 V: the monitor sinks at most 3 A.

Door servo: Feetech, via the FE−URT−2 board (as on the MES2 Harvester)

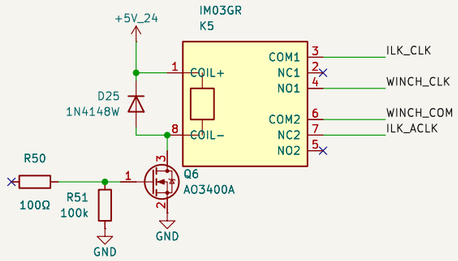


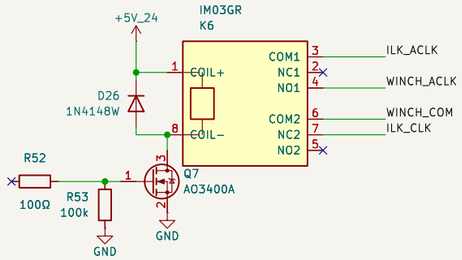
J15 SERVO 24V and J17 SERVO UART: same pinout
as CN5 / CN13 on the MES2 Harvester, so its cables fit.
The FE−URT−2 turns the UART into the Feetech half−duplex
servo bus and passes the 24 V on to the servos.
UART from the SC16IS740 I²C bridge (Winch
sheet, 0x48): TX → pin 4, pin 3 → RX.
F9 3 A PTC on the servo 24 V. F10 1.1 A PTC on the
FE−URT−2 5 V, fed from the screen converter so a
servo−side short can't pull down the module supply.
24 V only for 24 V servo types.
No module GPIO used (MES4 standard pin map).

Sheet 9: IMU 9 V supply (as on the MES3.1 carrier) + screen 5 V USB−C power
Aaron Raj Bhatti
Monkey Robotics
Sheet: /AuxPower/
File: AuxPower.kicad_sch
Title: MES4−Carrier−EC

Size: A3	Date: 2026−10−27	**Rev: A**
KiCad E.D.A. 9.0.2		Id: 9/10

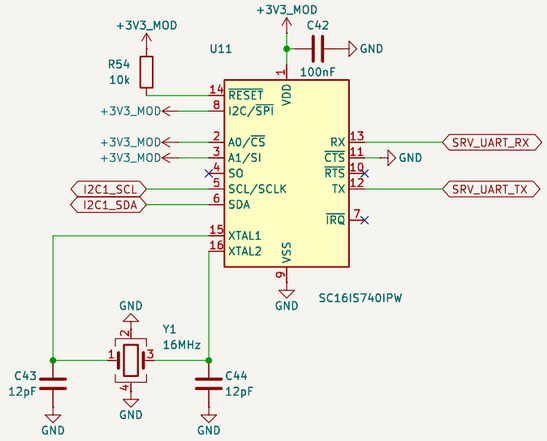
Winch direction relays (as U1 / U2 on the MES2_Extension)







Servo UART: SC16IS740 I²C–UART bridge (I²C1 0x48)



U11 NXP SC16IS740IPW: I²C1 slave at 0x48 (A1 = A0 = VDD; INA226s 0x40–0x45, TCA9535 0x20).
TX / RX → servo UART terminal (AuxPower sheet, 100 Ω + ESD there) → FE-URT-2.
Y1 16 MHz (CL 9 pF) + C43 / C44 12 pF: baud = 16 MHz / (16 × divisor):
1 Mbit/s (Feetech default, divisor 1), 500k (2), 250k (4) are exact.
IRQ not connected (no module pin free): poll the LSR / RXLVL registers.
CTS tied low (always clear), RTS open, RESET pulled up (soft reset via IOControl).
Uses no module GPIO, so the MES4 standard pin map stays intact.

Notes

Same relays and terminal as the MES2_Extension "WINCH CONTROL": TE IM03GR, 5 V coil (178 Ω, 28 mA) from +5V_24, contacts 2 A / 220 VDC, 60 W DC. They only switch the 48 V winch controller's direction inputs to J18 pin 2 (COM): pin 1 = CLK, pin 3 = ACLK. They are not rated for the winch motor current.
TE terminal assignment: coil 1 (+) / 8 (−); pole 1: 3 COM, 2 NC, 4 NO; pole 2: 6 COM, 7 NC, 5 NO.

Drivers as on the relay sheets (CIF §7a): AO3400A, 100 Ω gate, 100k pull-down, so both relays are OFF during reset and boot (the gate inputs float then). The Extension used 2SC2884 transistors without a pull-down.

Hardware interlock (new, uses the spare second pole): CLK closes COM → ILK_CLK → K5 NO1 → CLK only while K6 rests (its NC2 feeds ILK_CLK), and the same the other way. If both relays are energised, both outputs stay open, so the winch can never see both directions at once. Firmware still drives only one.

TODO (HANDOFF.md step 2): the gate resistors are not driven yet. PA0 / PA1 went back to relay coil feedback (MES4 standard); the winch moves to TCA9535 outputs + KAQY214S photoMOS relays.

Sheet 10: winch direction relays (IM03GR, as on the MES2_Extension) with hardware interlock
Aaron Raj Bhatti
Monkey Robotics
Sheet: /Winch/
File: Winch.kicad_sch

Title: MES4-Carrier-EC

Size: A3	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 9.0.2		Id: 10/10